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Tectonic implications of remagnetized upper Paleozoic strata of the northern Sierra Nevada

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ABSTRACT

A paleomagnetic study of a remagnetized Paleozoic terrane in the northern Sierra Nevada has provided new data on Cenozoic tectonics in the western United States. A combination of paleomagnetic and geologic evidence establishes that the remagnetization occurred during Late Jurassic time and that the rocks have experienced no major tectonic movement since. This conclusion establishes an eastern limit for the extent of large-scale tectonic rotations recently reported for Tertiary units on the western edge of the North American continent.

INTRODUCTION

Upper Paleozoic arc volcanic and volcaniclastic rocks are exposed in the northern Sierra Nevada in a 120-km-long belt extending from Lake Almanor to the North Fork of the American River (Fig. 1). The sequence includes Upper Devonian siliceous to mafic calc-alkalic volcanic rocks (Sierra Buttes, Elwell, and Taylor Formations); Carboniferous chert, siltstone, and intraformational breccia (Peale Formation); and Permian mafic and intermediate volcanic rocks (Goodhue and Reeve Formations) (D'Allura and others, 1977). Folding and thrusting during the Late Jurassic Nevadan orogeny produced near-vertical, north-to-northwest–striking packages of strata, bounded by faults. These strata were subsequently intruded by Jurassic to Cretaceous granitic plutons (Evernden and Kistler, 1970) and were partly covered by Tertiary volcanic rocks (Durrell, 1966). Neogene block faulting has produced the present basin and range topography.

SITE SELECTION

Samples suitable for paleomagnetic study were selected from the Permian Reeve Formation and the Upper Devonian Taylor Formation. The Reeve Formation includes pyroxene- and plagioclase-rich andesitic flows, shallow intrusions, and penecontemporaneous volcaniclastic turbidites. Three widely separated fossil localities indicate a middle Permian (Leonardian to Guadalupian) age for the turbidite sequences (D'Allura, 1977, and unpub. data). Samples were taken from the fine-grained tops of turbidites in three separate basins (Fig. 1). One of these, a 250-m-thick section near Homer Lake was sampled at three sites, at the base, middle, and top of the section. The volcaniclastic turbidites are extremely immature, comprising broken pyroxene, plagioclase, and magnetite crystals and dark volcanic lithic fragments. Metamorphism is weak; plagioclase grains are slightly clouded with saussurite or sericite, and glassy fragments are partially recrystallized to chlorite, epidote, and albite, but pyroxene grains are unaltered. Examination of polished sections reveals maghemite rims on the large magnetite grains, and very fine magnetite and maghemite disseminated in glassy lithic fragments.

The Taylor Formation includes andesitic tuff, breccia, flows, shallow intrusive rocks, and sparse volcanogenic sedimentary rocks. The Taylor has not been dated directly, but it is bracketed by Late Devonian and Early Mississippian ages. The Taylor conformably overlies the Elwell Formation, which contains Late Devonian ammonoids in quartzite (Anderson and others, 1974) and radiolaria in phosphatic chert (DeVay and Stanley, 1979). The Taylor is overlain by the Peale Formation, which has yielded an Early Mississippian brachiopod fauna from its upper member (McMath, 1966). Samples were collected from a sequence of interbedded flows and volcanogenic sedimentary rocks on Keddie Ridge and two sections of tuff, sedimentary rocks, and sparse flows along Indian Creek (Fig. 1). Plagioclase phenocrysts are thoroughly albitized or saussuritized, but pyroxene grains are generally unaltered. The formerly glassy groundmass and lithic fragments are altered to a fine mosaic of quartz, feldspar, and chlorite. Minor pyrite is the only opaque mineral visible in polished section. The magnetic minerals are either very fine grained, below the limit of resolution of the microscope, or very sparse. The average intensity of magnetization is two orders of magnitude lower than that of the Reeve. Variations in intensity during heating or alternating field demagnetization, however, are similar, suggesting similar magnetic mineralogy.

Although the lack of small-scale folding prohibits sampling of both limbs of a

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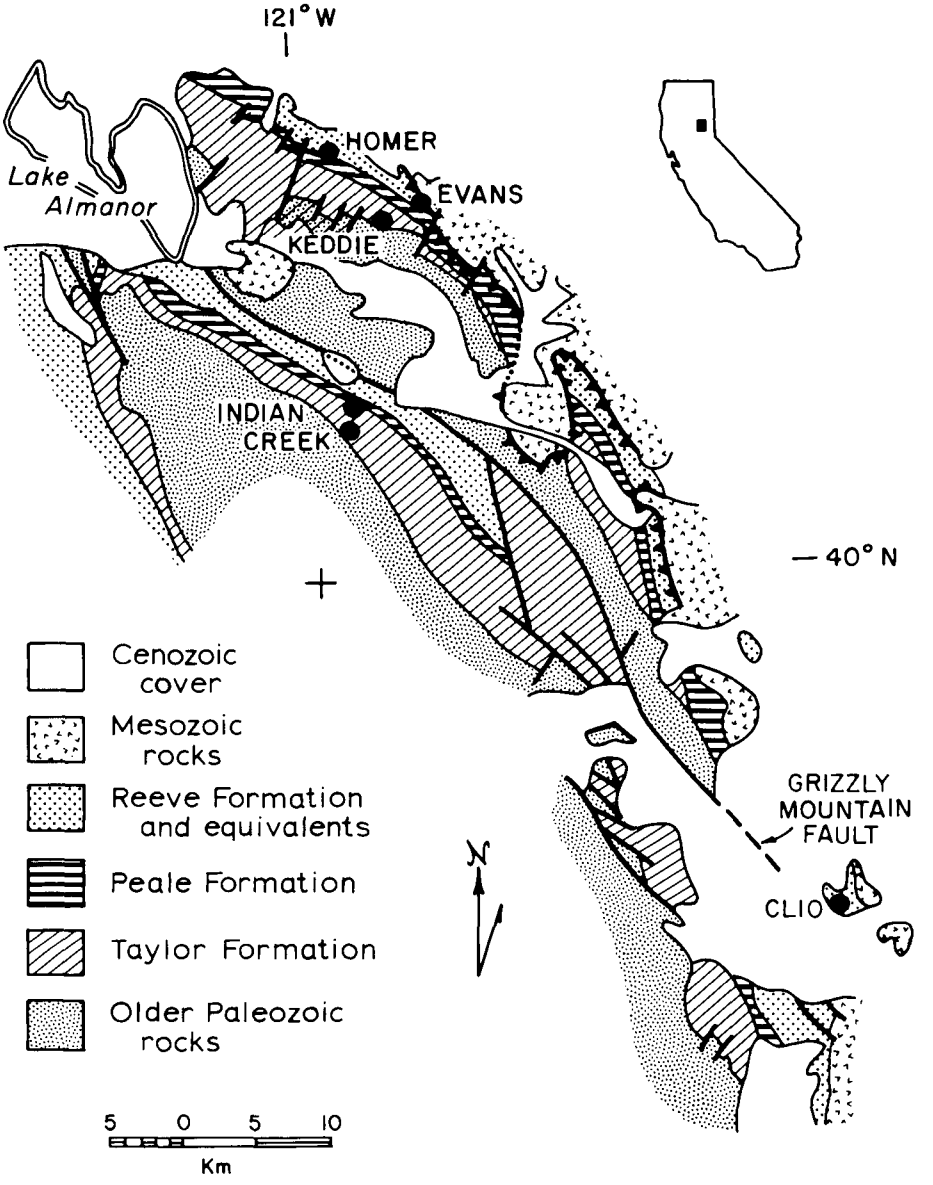


Figure 1. General geology of Paleozoic rocks in northern Sierra Nevada; circles indicate paleomagnetic sample sites. Oroclinal flexure is indicated by change in strike of contacts between Clio and Lake Almanor.

single fold, sample sites were chosen to test the effects of various structural features on the measured paleomagnetic directions. For example, the Taylor Formation was sampled on both sides of the Grizzly Mountain Fault, a major pre-Tertiary fault that repeats the section. In addition, samples from the Reeve Formation were taken from both ends of an oroclinal flexure of unknown age which has produced a northward change in the strike of bedding from about N10°W to N60°W. The bedding attitudes are generally uniform within structural blocks.

RESULTS

Analysis of the samples included stepwise alternating field (af) demagnetization in peak fields up to 600 Oe. Typically, less than half of the original remanent magnetism remains after demagnetization at 300 Oe, and the samples display no significant change in vector orientation with subsequent demagnetizations. Vector orientations in most samples stabilize in peak fields as low as 150 Oe. Except for variable, soft, viscous remanence, only one component of magnetization is recognized (Fig. 2).

Site and formation means and α_{95} values, without correction for tilt of bedding, are listed in Table 1. Fifty-two samples from

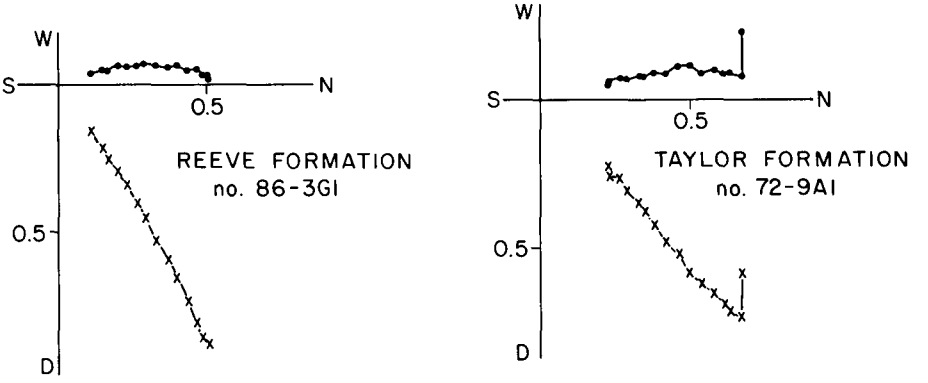


Figure 2. Effects of alternating field demagnetization on direction of magnetic vector, plotted by method of Zijderveld (1967). Crosses represent projections onto north-south vertical plane; circles represent projections onto horizontal plane.

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TABLE 1. SITE AND FORMATION MEANS OF MAGNETIC VECTOR DIRECTIONS

Site	N	Without tilt correction I	D	α_{95}	κ	With tilt correction I	D	α_{95}	κ
Reeve Formation									
Homer A	14	38.1	319.2	21.9	4.3	6.8	341.6	21.9	4.3
Homer B	12	51.6	341.3	6.0	53.2	5.2	356.1	6.0	53.2
Homer C	5	60.4	314.8	19.0	17.2	18.2	359.6	20.9	14.3
Evans	8	51.4	352.3	9.0	39.0	-3.2	7.9	9.7	33.5
Clio	13	62.3	8.2	6.2	46.2	17.4	76.0	5.5	57.8
All Reeve	52	53.8	339.8	6.8	9.3	10.7	13.1	11.6	3.9
Taylor Formation									
Keddie Ridge	22	50.0	350.3	5.0	38.8	-18.8	1.1	5.8	29.7
Indian Creek A	20	67.0	310.7	10.1	11.5	16.7	244.1	10.0	11.7
Indian Creek B	11	60.2	299.1	14.3	11.1	13.4	251.8	15.5	9.7
All Taylor	53	60.3	329.6	6.0	11.5	1.5	290.9	19.9	2.0

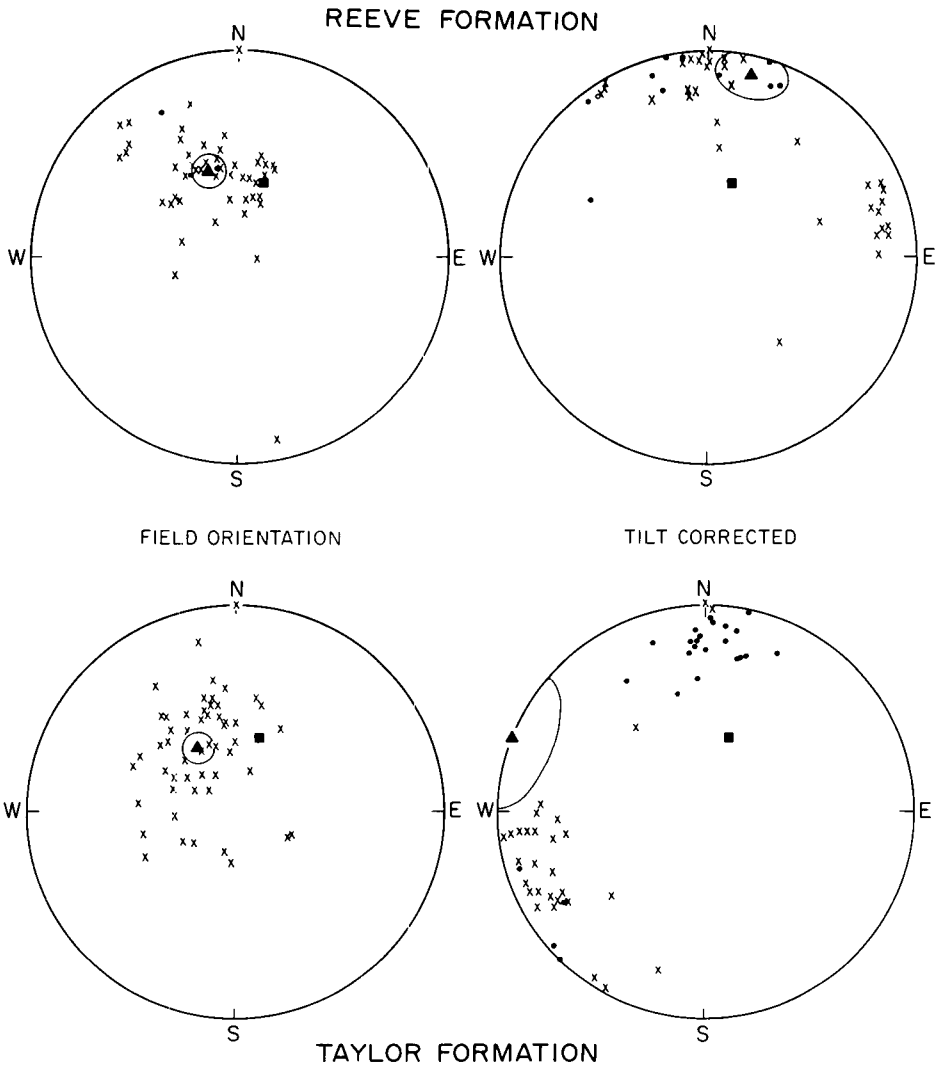


Figure 3. Equal-area stereographic projections of paleomagnetic vectors for Reeve and Taylor Formations before and after correction for tilt of bedding. Crosses indicate positive inclinations; circles indicate negative inclinations; squares indicate present field direction; triangles indicate mean paleomagnetic direction with projection of cone of 95% confidence.

the Reeve Formation cluster about a mean direction of $D = 340°$, $I = 54°$, $\alpha_{95} = 7°$, corresponding to a pole position of 136°E, 73°N. Fifty-three samples from the Taylor Formation yield a mean direction of $D = 330°$, $I = 60°$, $\alpha_{95} = 6°$, corresponding to a pole position of 162°E, 67°N (Fig. 3). A simple F test indicates that the two paleomagnetic pole positions are statistically indistinguishable. Each direction is, however, distinct from the present field direction at the 95% confidence level. The consistency of the magnetic directions from the two formations, separated in age by 100 m.y. and both complexly deformed, strongly suggests a post-Permian remagnetization. One means of testing this hypothesis is to apply the paleomagnetic fold test by correcting for the tilt of the bedding. This approach considerably increases the scatter of the data (Fig. 3; Table 1). Because there is little variation in strike and dip within each tectonic block, most of the increased scatter can be attributed to variations in strike and dip between tectonic blocks. For example, the bimodal distribution of tilt-corrected data from the Taylor Formation reflects sample sites on either side of the Grizzly Mountain fault. Strictly speaking, the fold test should be applied only to data for which both limbs of the same fold may be sampled. In the present situation, use of the fold test presupposes that the observed tilts result from a single, simple rotation about the strike, and it thus neglects the actual, complex tectonic history of these rocks.

These difficulties do not affect the determination of the angle of magnetic inclination with respect to bedding. An alternate approach is to examine the distribution of inclinations (Fig. 4). In each formation, the scatter is comparable for the uncorrected inclinations and the tilt-corrected inclinations, and the test is inconclusive. Thus, on the basis of an analysis of the paleomagnetism alone, it is not possible to confirm the remagnetization hypothesis.

Several lines of geological and geophysical evidence, however, do support the idea of a postfolding remagnetization. First, the paleomagnetic poles, calculated without tilt correction, fall on the North American apparent polar wander path at 140 to 150 m.y. (Fig. 5)—a remarkable coincidence if it is not due to Late Jurassic remagnetization. Second, the proposed timing of the Nevadan orogeny agrees well with this Late Jurassic age and would

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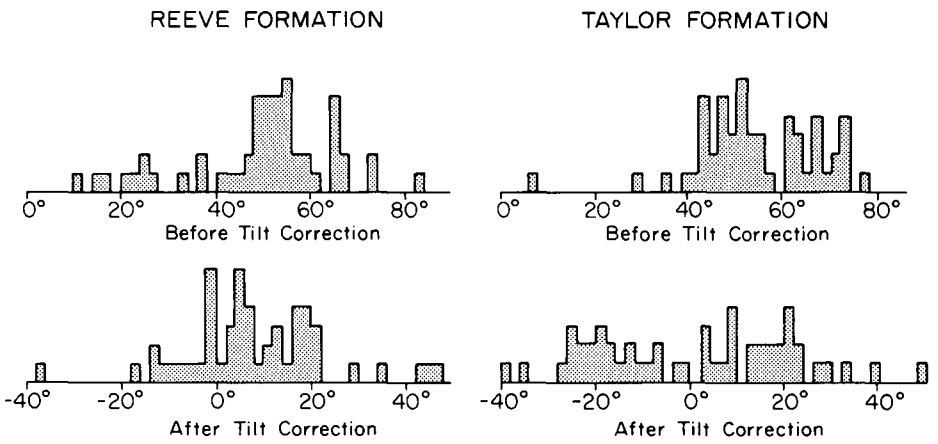


Figure 4. Histograms of paleomagnetic inclinations for Reeve and Taylor Formations before and after correction for tilt of bedding.

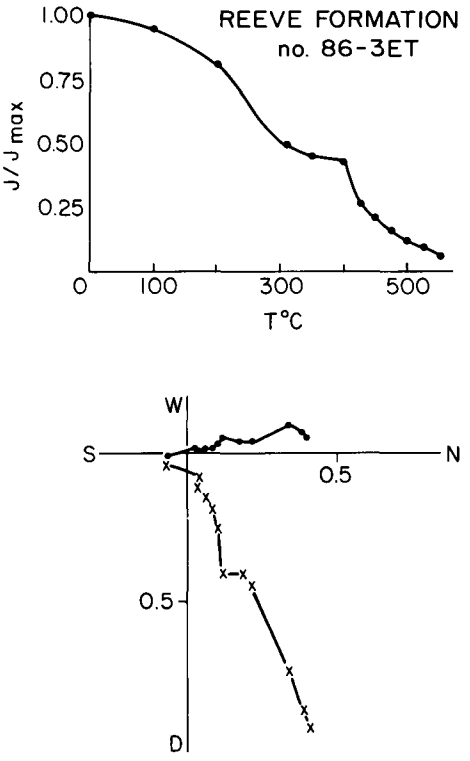


Figure 6. Effects of thermal demagnetization on intensity and direction of magnetic vector, plotted by method of Zijderveld (1967). Symbols as in Figure 2.

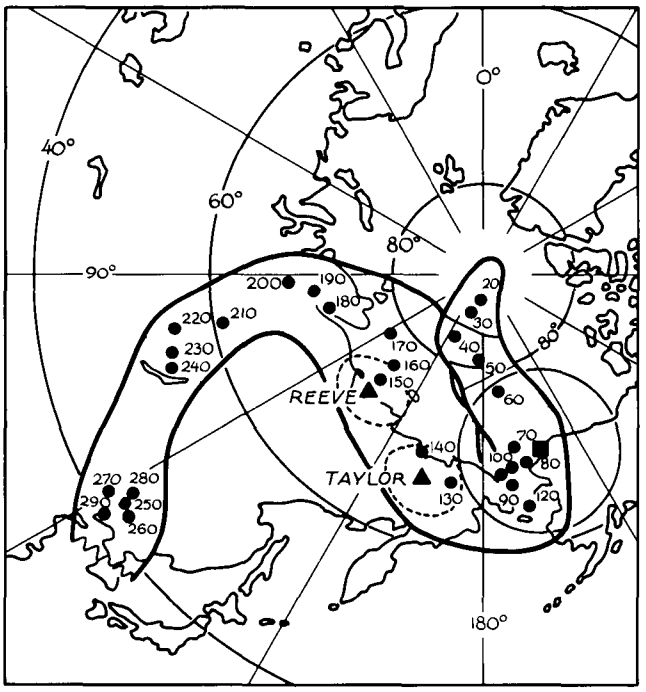


Figure 5. Apparent polar wander path for North American craton, after Irving (1979). Circles represent mean paleomagnetic pole positions at 10-m.y. intervals, and envelope represents standard error about those means. Triangles indicate paleomagnetic pole positions calculated for 52 samples from Permian Reeve Formation and 53 samples from Devonian Taylor Formation, with their 95% confidence ovals. Square indicates mean pole position and associated A_{95} for Late Cretaceous plutons of Sierra Nevada (Grommé and Merrill, 1965).

provide a mechanism for the remagnetization. The Taylor and Reeve Formations display a northwest-trending penetrative deformation and are cut by undeformed plutons. Although ages are not yet available for these plutons, K-Ar ages from granitic rocks of surrounding areas are all Cretaceous (Evernden and Kistler, 1970). The entire upper Paleozoic sequence is thrust eastward over similarly foliated Callovian (upper Middle Jurassic) volcanic rocks (McMath, 1966). Callovian and Kimmeridgian rocks to the west are affected by a parallel foliation and are cut by 136- to 140-m.y.-old granitic intrusions (Moores and others, 1979). Finally, in spite of the wide stratigraphic distribution of sites within the Reeve Formation, all of the tilt-corrected magnetic directions represent normal polarity. In view of the preponderance of reversed polarity states during Permian time (Irving, 1964), such a result would be quite unexpected if the magnetization were, in fact, primary.

We therefore conclude that the magnetization of the Reeve and Taylor Formations is a secondary magnetization acquired during Late Jurassic time. The Late Jurassic orogenic event evidently produced enough secondary magnetite and maghemite, by alteration of volcanic glass, to obliterate the primary magnetization. Twelve samples, six from each formation, were thermally demagnetized at successive temperatures in an effort to remove the secondary magnetization. Intensities fall to 5%–10% of the original natural remanent magnetization at 550 °C (Fig. 6). Samples from the Reeve Formation yield very stable directions consistent with results from af demagnetization. Directions from the Taylor samples are widely scattered. Breakdown of chlorite at elevated temperatures may have formed new iron oxides that acquired a weak partial thermal remanent magnetization due to an imperfectly nulled field present during heating. As the natural remanent magnetization of the Taylor samples is low (averaging about 10^{-6} emu/cm^3), even a very weak partial thermal remanent magnetization could significantly affect the vector orientation. Thus, thermal demagnetization failed to uncover the pre-Jurassic magnetization.

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IMPLICATIONS

Perhaps the most important implication of our work is its bearing on the nature of Cenozoic tectonics in the western United States. Several studies have shown that Mesozoic and lower Tertiary units along the western edge of the North American continent have been rotated clockwise by amounts ranging from 40° to 70° (Beck, 1976; Simpson and Cox, 1977; Beck and Burr, 1979). One explanation for these large tectonic rotations is that they arise from complex tectonic interactions associated with the subduction of the Kula and Farallon plates beneath the North American plate (Simpson and Cox, 1977). An alternate hypothesis is that the rotations are a manifestation of the opening of the Basin and Range province (Heptonstall, 1977). The circles of confidence associated with our data and with the presumed Jurassic pole position allow us to conclude that the northern Sierra Nevada has experienced at most 10° of rotation (or north-south translation) since Jurassic time. Any movement of the northern Sierra Nevada associated with extension of the Basin and Range province is, therefore, restricted to east-west translation, resulting from rotation about a pole very close to the paleomagnetic pole. Thus, the northern Sierra Nevada has not participated in the microplate rotations of units situated to the northwest, and the driving force for the microplate motions is probably not the extension of the Basin and Range province to the east and southeast of the northern Sierra Nevada. In view of our results and those of an earlier paleomagnetic study of the southern Sierra Nevada (Grommé and Merrill, 1965), it is likely that the entire Sierra Nevadan block, from the deformed Paleozoic terrane in the north to the Mesozoic batholith in the south, has behaved as a single tectonic unit and has experienced no significant rotations at least since Cretaceous time.

CONCLUSIONS

The geological and paleomagnetic data described above support the hypothesis of a complete remagnetization of the Reeve and Taylor Formations during Late Jurassic time, due to the formation of secondary iron oxides. The excellent fit with the North American apparent polar wander path implies little or no movement of these units relative to North America since the Late Jurassic. The following conclusions may be drawn.

The upper Paleozoic arc terrane of the northern Sierra Nevada was in place on the western margin of North America by Late Jurassic time. The coincidence of the paleomagnetic poles of the Reeve and Taylor Formations with the Late Jurassic North American pole would be remarkable if the arc terrane were located elsewhere.

The present metamorphic mineral assemblage was established during Late Jurassic regional metamorphism. Iron oxides, the magnetic carriers, occur as fine grains intergrown with chlorite; both must have formed during the Jurassic to reset the pole positions.

The westward oroclinal flexure at the north end of the Sierra Nevada formed prior to the Late Jurassic remagnetization. In the Reeve Formation, the differences in mean declinations between the Clio site and the northern Homer and Evans sites are far smaller than the 50° change in strike of bedding. The bend in the regional strike, therefore, must predate the Late Jurassic remagnetization.

The Sierra Nevada block has not been involved in Cenozoic microplate rotations. If Cenozoic plate interactions rotated the Paleozoic section, the locations of pole positions for Taylor and Reeve strata on the North American apparent polar wander path would be the results of fortuitous coincidences.

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